AI CHATBOT FOR COLLEGE

FACULTY REVISED SYSTEM ARCHITECTURE & DIAGRAM DOSSIER

✓ 3 Swimlanes (Student | Chatbot | Database) ✓ Database Student Record Fetching ✓ 4-Tier Attendance Detention Policy

1. Activity Diagram — 3-Swimlane Session Flow with Database Integration

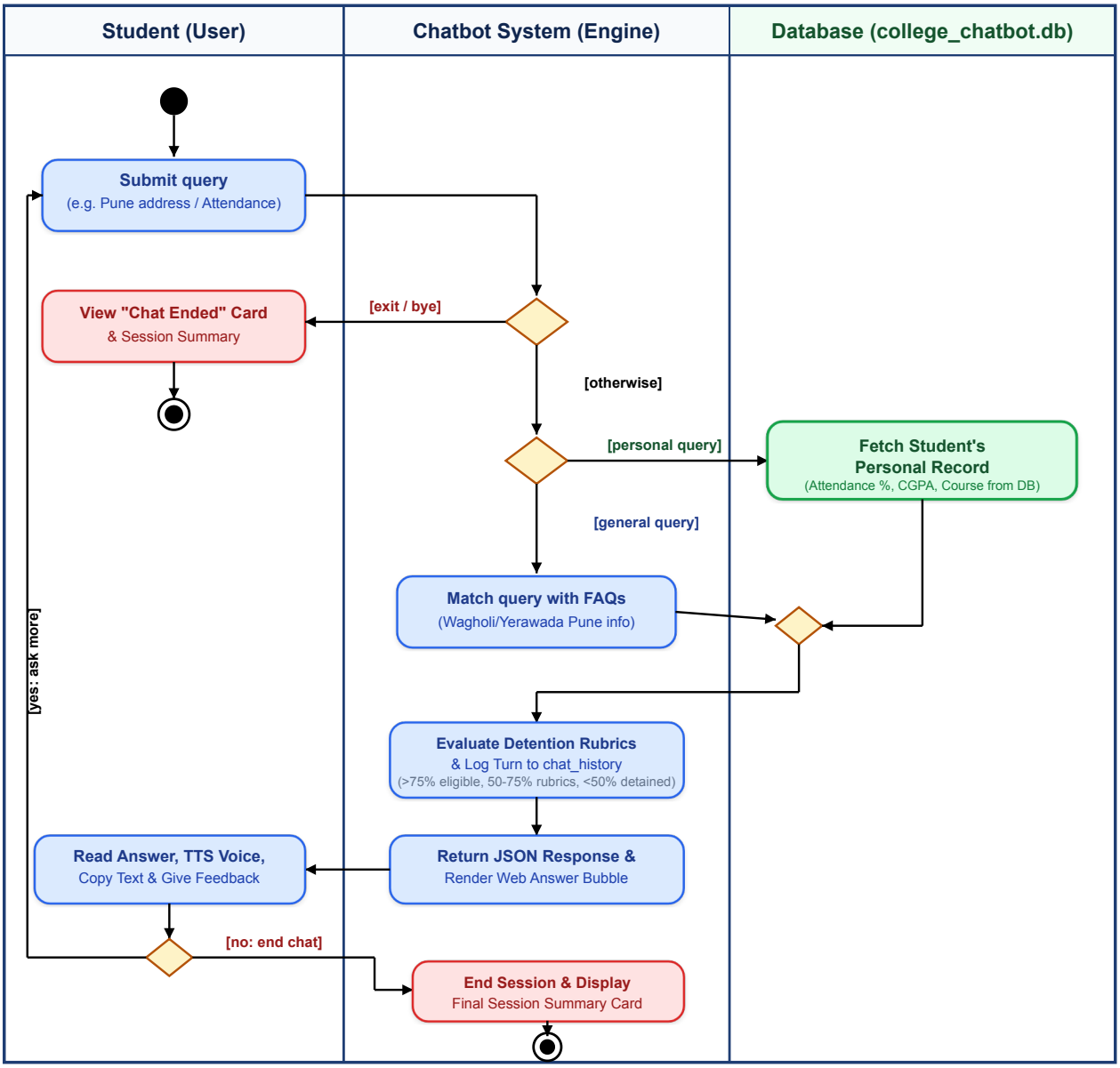


Fig 1: Activity Diagram — 3-Swimlane architecture (Student, Chatbot System, Database) showing explicit student personal record retrieval (attendance & CGPA), FAQ matching for Pune campuses (Wagholi & Yerawada), 4-tier detention policy evaluation, and controlled session loop with explicit End states.

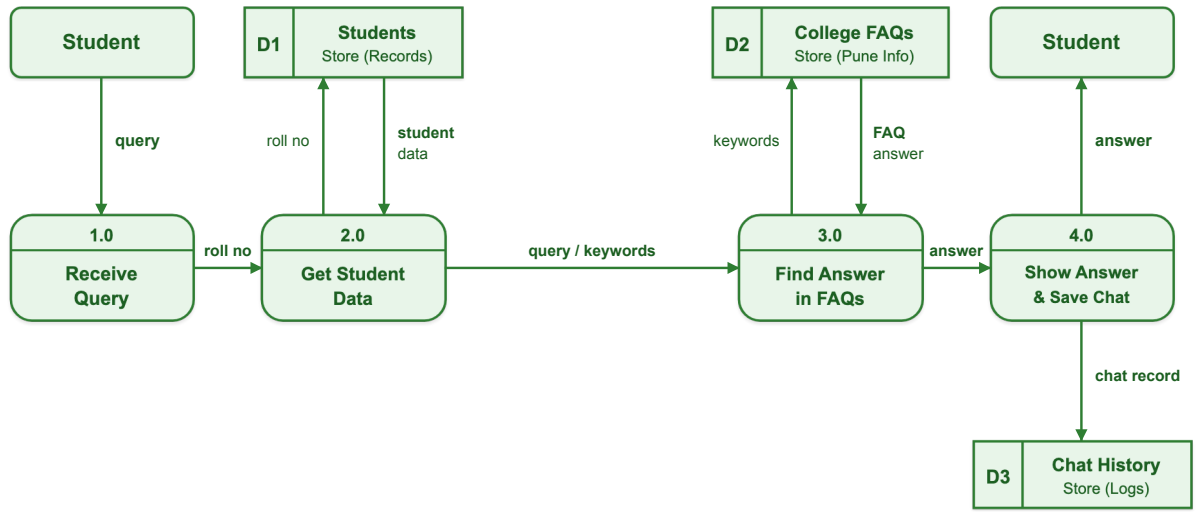
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✓ Level 1 Process Decomposition (1.0 – 4.0) | ✓ 3 Data Stores (D1 Students, D2 FAQs, D3 Chat) | ✓ Official DFD Symbols Reference Table

2. Data Flow Diagram (DFD Level 1) — Detailed Process Flows & Symbols Table

Data Flow Diagram (Level 1): AI Chatbot for College



DFD symbols and their uses

Symbol	Name	Use	Example in our project
Student	**External entity**	A person or external system who provides or receives data	**Student (User asking questions)** e.g. Pune campus location, attendance check
1.0 Process	**Process**	A transformation step that operates on data (receives, calculates, finds)	**3.0 Find Answer in FAQs** Evaluates Wagholi/Yerawada & detention rubrics
D1 Store	**Data store**	A repository where persistent data is stored and retrieved	**D2 College FAQs Store & D1 Students** college_chatbot.db (SQLite database)
→	**Data flow**	A directed arrow showing data moving between entities, processes and stores	**query, answer, roll no, student data** Directional data exchange across all 4 steps

Fig 2: Data Flow Diagram (DFD) — First-Level Process Decomposition (1.0 Receive Query, 2.0 Get Student Data, 3.0 Find Answer in FAQs, 4.0 Show Answer & Save Chat) with data stores (D1 Students, D2 FAQs containing Pune Wagholi & Yerawada info, D3 Chat History) and formal academic DFD symbols reference table.

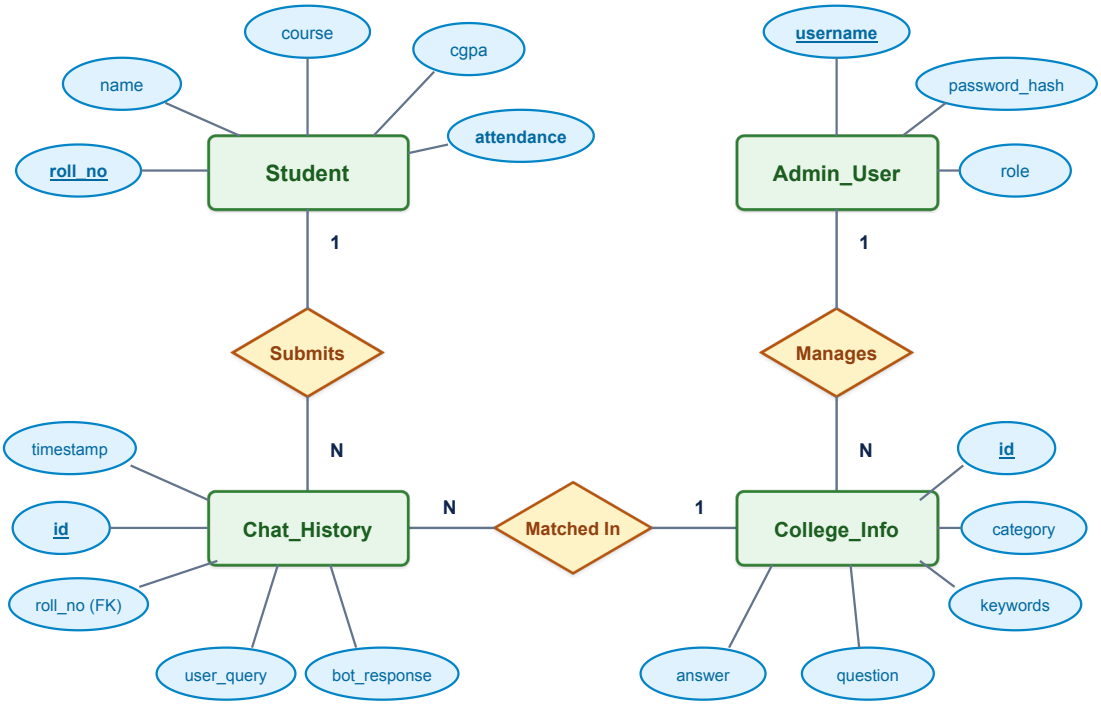
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✓ Classic Academic ER Notation | ✓ 4 Core Entities (Student, Admin, Chat, Info) | ✓ 3NF Normalization Defense & Legend Key

3. Entity-Relationship Diagram (ERD — Classic Notation) & Relational Schema

ER Diagram (Classic Notation): AI Chatbot for College



Schema & Project Knowledge Base Defense:

- **Student Entity:** Stores attendance %, CGPA & evaluates 4-tier detention policy.
- **College_Info Entity:** Contains Pune campuses (Wagholi & Yerawada) & exam rules.
- **Chat_History Entity:** Logs each turn with student roll_no FK, query & bot answer.
- **Admin_User Entity:** RBAC authentication for updating college FAQs & detention data.

Verified in 3NF Normalization: Zero transitive dependencies & lossless join preservation.

Key / Legend:

- Entity (Table)
- Attribute (Column) — id = Primary Key
- Relationship
- Link / Cardinality (1:N)

Fig 3: Entity-Relationship Diagram (ERD — Classic Notation) — 4 core entities (Student, Admin_User, Chat_History, College_Info) with attributes (PK underlined, FK indicated), cardinalities (1:N, N:1), and official symbol key matching faculty review specifications.